Informal Summary of Theorem 3.2

Theorem 3.2 says that if a watermark changes how the model samples text, then it also leaves a detectable statistical trace. That trace grows with the length of the generated text, so longer outputs are easier to distinguish from ordinary model samples. The effect is strongest for greedy and biased schemes, weaker for semantic schemes, and absent only for distribution-preserving schemes.

Informal Summary of Theorem 3.4

Theorem 3.4 says that editing makes watermark verification harder by removing part of the watermark signal. For token-level methods, this loss depends directly on the number of edited tokens. For semantic methods, it depends on how often editing changes the sentence-level watermark evidence. If too much signal is lost, reliable verification becomes impossible. This is why token-level methods perform best under light editing, whereas semantic methods are better when wording changes substantially while the sentence-level signal remains intact.

Informal Summary of Theorem 4.2

Theorem 4.2 states that the best watermark family is the feasible one that requires the smallest information budget to meet the robustness target. If a distribution-preserving scheme is sufficiently robust, it is optimal because it achieves the target without statistical drift. Otherwise, the selector compares the required budgets for token-level and semantic watermarking and selects the smaller feasible option. If neither is feasible, then the desired operating point cannot be achieved within this family class.